

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Summer 2026 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVC601 - DESIGN OF RC STRUCTURES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1. The modular ratio (m) used in the Working Stress Method for M25 grade concrete is approximately:
☒ a. 10.98
☐ b. 13.33
☐ c. 18.66
☐ d. 20.00
2. For a balanced singly reinforced rectangular section using Fe500 steel under LSM, the limiting depth of the neutral axis ($x_{u,max}/d$) is
☐ a. 0.53
☐ b. 0.48
☒ c. 0.46
☐ d. 0.39
3. In a doubly reinforced concrete beam, compression steel is primarily added to:
☐ a. Increase the shear capacity of the support
☐ b. Resist excessive bending moments when beam dimensions are architecturally restricted
☐ c. Prevent torsional cracking along the span
☐ d. Reduce the structural nominal cover constraints
4. The minimum clear cover for a standard RCC beam to satisfy durability requirements under 'moderate' exposure conditions should be
☐ a. 20 mm
☐ b. 25 mm
☐ c. 30 mm
☐ d. 45 mm
5. What is the absolute maximum spacing permitted for horizontal shear reinforcement (stirrups) along the axis of an RCC beam?
☐ a. $0.75d$ or 300 mm, whichever is smaller
☐ b. $1.0d$ or 450 mm, whichever is smaller
☐ c. $3d$ or 450 mm, whichever is smaller
☐ d. $0.5d$ or 150 mm, whichever is smaller

- 6 The maximum diameter of any reinforcing steel bar embedded in an RCC slab must not exceed:
 - a. One-fourth of the total slab thickness
 - b. One-eighth of the total slab thickness
 - c. One-half of the total slab thickness
 - d. 16 mm regardless of thickness
- 7 A heavily reinforced concrete column is designed with high-strength Fe500 steel longitudinal bars. In the Limit State of Collapse under pure axial compression, what is the maximum design stress allowed in these steel bars?
 - a. $0.87f_y$
 - b. $0.75f_y$
 - c. $0.67f_y$
 - d. $0.446f_y$
- 8 When designing a stair flight slab that is supported structurally at the top and bottom risers by walls parallel to the risers, what is the effective span length according to IS 456?
 - a. The horizontal distance between the centerlines of the supporting walls.
 - b. The total inclined length of the soffit slab.
 - c. The clear horizontal distance between the supports plus the horizontal width of one tread.
 - d. The horizontal distance between the centers of the supports plus half the width of the landing at each end.
- 9 If an RC column has an unsupported length of 4.0 m and a cross-section of 300mmx400 mm, what is its computed minimum design eccentricity e_{min} along the major axis 400 mm dimension)?
 - a. 20.0 mm
 - b. 21.3 mm
 - c. 23.3 mm
 - d. 25.0 mm
- 10 Why does the code restrict the maximum spacing of shear stirrups to 300 mm even if calculations for low shear forces indicate a much larger required spacing?
 - a. To support the weight of the main longitudinal reinforcement cage during casting.
 - b. To ensure that any potential diagonal shear crack developing at 45° will be intercepted by at least one stirrup.
 - c. To mitigate the localized effects of differential concrete shrinkage.
 - d. To maintain the minimum structural ductility index required under extreme wind loading.
- 11 While designing an isolated shallow square footing, critical section for checking one-way (shear) action is located at:
 - a. At the face of the column
 - b. At a distance of 'd' from the column face
 - c. At a distance of 'd/2' from the column face
 - d. Along the diagonal center-line of the pad
- 12 For checking punching shear (two-way action) in an isolated footing slab, the critical shear perimeter is checked at:
 - a. At the face of the column
 - b. At a distance of 'd' from the column face
 - c. At a distance of 'd/2' from the column face
 - d. At a distance of '2d' from the column face

Q2. Solve the following.

- A) Compare working stress method & limit state method of design

- B) Calculate the moment of resistance and safe udl that the beam can carry by WSM for the following details: Effective simply supported span = 4.5 m, $b = 230$ mm, $D = 450$ mm, Area of steel – 4 bars of 25 mm diameter on tension side and 3 bars of 16 mm diameter on compression side, Effective cover = 40mm. Material M20 and Fe415. $\sigma_{cbc} = 7$ N/mm², $\sigma_{st} = 230$ N/mm², $m = 13.33$. 6
- Q3. Solve the following.
- A) Explain balanced, under reinforced and over reinforced section in RC beam 6
- B) Singly reinforced beam 250 mm × 500 mm in section is reinforced with 4 bars of 16 mm diameter with an effective cover of 25 mm. Effective span of the beam is 5 m. Assuming M20 concrete and Fe 415 steel, determine the central concentrated load P that can be carried by the beam in addition to its self-weight. Use LSM. 6
- Q4. Solve Any Two of the following.
- A) A rectangular Beam is to be simply supported on supports of 230 mm width. The clear span of the beam is 6m. The Beam is to have a width of 300 mm. The characteristics superimposed load is 12 kN/m. Using M20 concrete and Fe 415 steel, design the beam. 6
- B) A simply supported beam 300mm x 600mm effective is reinforced with 5 bars of 25 mm diameter. It carries a udl of 80 kN/m including self-weight, over an effective span of 6 m. Out of the 5 main bars, 2 bars can be bent up safely near the supports. Design the shear reinforcement for the beam. Use M20 grade of concrete and Fe415 steel. 6
- C) Explain in brief- a) Development length, b) types of bond's 6
- Q5. Solve Any Two of the following.
- A) Design a simply supported RCC slab over a room of size 2.5m x 7m. The thickness of supporting wall is 230 mm and the slab carries 75 mm lime concrete on its top, the unit weight of which is 20kN/m³. The live load on slab may be taken as 2kN/m². Use M20, Fe415 and LSM. Apply deflection check only. 6
- B) Design a slab over a room having dimensions 4mx5m. The edges of the slab are simply supported and the corners are held down. The LL on the slab is 2.5 kN/m². The slab has bearing of 300 mm on the supporting walls. Use M15 concrete and Fe415 steel. 6
- C) The stair of residential building has a located in a stair measuring 2.5m x 4 m. The vertical distance between the floors is 3.20 m. Design the dog legged stair. The live load on stair is 2.5 kN/m². Use M15 concrete and Fe415. 6
- Q6. Solve Any Two of the following.
- A) Find working Load carrying capacity of an axially loaded tied column with an unsupported length of 4.6 m. The column is fixed at one end and pinned at the other end. column is 400 mm x 500 mm in size and reinforced with 6 Nos of 16 mm bars and use M 20 & Fe415. 6
- B) Explain longitudinal and transverse reinforcement for columns. 6
- C) Design a square footing of uniform thickness for an axially a loaded column of (300mm x 300mm) size. The SBC of soil = 250kN/m² Column carries a axial load of 1200kN. Use M25 concrete and Fe500 steel. Also draw reinforcement details. Use LSM. 6

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